

Exponential Families

A Crash Course from First Principles

Parametric Inference, B.Stat. 3rd Year — Lectures 4–5, and everywhere else

Fourth of the crash courses. This one is different in kind: Bayes, testing and minimax are *topics*, but exponential families are the *structure underneath all of them*. Read this and a lot of the course stops looking like a list of separate results.

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1 Why this class of distributions exists

Notice something about the whole course. The same handful of families keeps appearing — Binomial, Poisson, Exponential, Normal, Gamma — and for all of them *everything works*:

- a low-dimensional sufficient statistic exists,
- it turns out to be complete, so Lehmann–Scheffé applies,
- the Cramér–Rao bound is attained,
- the family has MLR, so UMP tests exist,
- there is a tidy conjugate prior with two hyperparameters.

Meanwhile two families keep breaking things: the **Cauchy** (sufficient statistic is the whole order statistic, not complete, no closed-form MLE, no MLR, no useful conjugate prior) and $U[0, \theta]$ (Cramér–Rao meaningless, non-standard rates).

The single fact that explains all of this

The well-behaved families are all **exponential families** — the same object wearing different clothes. The badly behaved ones are not.

So exponential families are not one more topic. They are the *hypothesis* of most of the theorems in the course, and the two villains are villains precisely because they fail it. Everything in Section 5 below is a consequence of one definition.

2 The definition, and how to recognise one

Definition (one parameter)

$\{f(\cdot | \theta) : \theta \in \Theta\}$ is a **one-parameter exponential family** if

(i) the **support** $\{x : f(x | \theta) > 0\}$ does **not depend on** θ , and

(ii) $f(x | \theta) = p(\theta) h(x) \exp\{c(\theta) T(x)\}$

for some functions $p > 0$, $h > 0$, c , T .

Both conditions matter, and condition (i) is the one people skip.

The recognition drill

Take logs. You have an exponential family iff

$$\log f(x | \theta) = \underbrace{\log p(\theta)}_{\text{pure } \theta} + \underbrace{\log h(x)}_{\text{pure } x} + \underbrace{c(\theta) T(x)}_{\text{a product: one } \theta\text{-factor} \times \text{one } x\text{-factor}},$$

with a support that does not move. So: expand $\log f$, sort the terms into three piles, and check that every term mixing θ and x separates into a product.

Quick test for failure. If it is an exponential family then $\frac{\partial^2}{\partial \theta \partial x} \log f = c'(\theta) T'(x)$, which must factor into (function of θ) $\times$ (function of x). If the mixed partial does not factor, it is not an exponential family.

2.1 Worked identifications

Binomial, $X \sim \text{Bin}(m, \theta)$:

$$\binom{m}{x} \theta^x (1 - \theta)^{m-x} = \underbrace{(1 - \theta)^m}_{p(\theta)} \underbrace{\binom{m}{x}}_{h(x)} \exp \left\{ \underbrace{\log \frac{\theta}{1-\theta}}_{c(\theta)} \cdot \underbrace{x}_{T(x)} \right\}.$$

Support $\{0, \dots, m\}$, free of θ . ✓

Poisson: $\frac{e^{-\theta} \theta^x}{x!} = \underbrace{e^{-\theta}}_p \underbrace{\frac{1}{x!}}_h \exp \left\{ \underbrace{\log \theta}_c \cdot \underbrace{x}_T \right\}.$

Exponential (mean θ): $\frac{1}{\theta} e^{-x/\theta} = \underbrace{\theta^{-1}}_p \underbrace{\mathbb{I}[x > 0]}_h \exp \left\{ \underbrace{-\frac{1}{\theta}}_c \cdot \underbrace{x}_T \right\}.$ Support $(0, \infty)$, free of θ . ✓

$N(\theta, 1)$: $\frac{1}{\sqrt{2\pi}} e^{-(x-\theta)^2/2} = \underbrace{\frac{e^{-\theta^2/2}}{\sqrt{2\pi}}}_p \underbrace{e^{-x^2/2}}_h \exp \left\{ \underbrace{\theta}_c \cdot \underbrace{x}_T \right\}.$

Geometric, $f = \theta(1 - \theta)^x$ on $x = 0, 1, \dots$: $\underbrace{\theta}_p \cdot \underbrace{1}_h \exp \left\{ \underbrace{\log(1 - \theta)}_c \cdot \underbrace{x}_T \right\}.$

Negative binomial (the HW1 Q1 form), $f = \frac{\Gamma(\alpha+x)}{\Gamma(\alpha)x!} \theta^\alpha (1 - \theta)^x$: $p = \theta^\alpha$, $h = \frac{\Gamma(\alpha+x)}{\Gamma(\alpha)x!}$, $c = \log(1 - \theta)$, $T = x$.

Power series family, $f = \frac{a_x \theta^x}{A(\theta)}$: $p = 1/A(\theta)$, $h = a_x$, $c = \log \theta$, $T = x$. (So Bernoulli, Binomial, Poisson, Geometric and Negative Binomial are all power-series families, hence all exponential families — one observation covering five rows of the table.)

2.2 And the failures

$U[0, \theta]$. $f = \frac{1}{\theta} \mathbb{I}[0 \leq x \leq \theta]$. The functional form is fine, but the **support moves with θ** : condition (i) fails outright. Not an exponential family.

$U[\theta, \theta + 1]$. Same reason.

Cauchy(θ). The support is all of $\mathbb{R}$, so condition (i) holds — the failure is structural. Apply the mixed-partial test with $u = x - \theta$:

$$\log f = -\log \pi - \log(1 + u^2), \quad \frac{\partial}{\partial x} \log f = \frac{-2u}{1 + u^2}, \quad \frac{\partial^2}{\partial \theta \partial x} \log f = \frac{2(1 - u^2)}{(1 + u^2)^2}.$$

This vanishes exactly at $x = \theta \pm 1$ — zeros whose *position moves with θ* . A product $c'(\theta)T'(x)$ has its x -zeros at fixed places. Contradiction, so the Cauchy is not an exponential family.

Second proof, and the more memorable one: in an exponential family $\sum_i T(X_i)$ is sufficient for every n — a **one-dimensional** statistic no matter how much data you have. For the Cauchy the minimal sufficient statistic is the whole order statistic, whose dimension grows with n . So it cannot be an exponential family.

Common confusion: “exponential distribution” vs “exponential family”

Unrelated names. The *exponential distribution* $\frac{1}{\theta}e^{-x/\theta}$ is one member of the exponential *family* class — but so are the Binomial, Poisson, Normal and Gamma, none of which is “an exponential distribution”. The word refers to the $\exp\{c(\theta)T(x)\}$ *form*, not to any particular density.

Equally: having an $e^{(\cdot)}$ in the formula neither proves nor disproves membership. $U[0, \theta]$ has no exponential and is not a member; the Cauchy has no exponential and is also not a member — but for a completely different reason.

3 Natural parameter and natural parameter space

The parametrisation you are handed is arbitrary. The theory is cleanest in the parameter that appears *linearly in the exponent*.

Definitions

$\eta := c(\theta)$ is the **natural parameter**, and

$$\Gamma := \{c(\theta) : \theta \in \Theta\} \subseteq \mathbb{R}$$

is the **natural parameter space**. Rewriting the density in terms of η gives the *natural form*

$$f(x | \eta) = h(x) \exp\{\eta T(x) - A(\eta)\}, \quad A(\eta) = \log \int h(x) e^{\eta T(x)} dx,$$

where A (the *log-partition* or *cumulant* function) is just whatever makes the density integrate to 1.

Common confusion: Θ versus Γ

The completeness criterion is a condition on Γ , **not** on Θ . Θ is almost always a nice interval; that tells you nothing. You must compute the image $c(\Theta)$.

Family	Θ	$c(\theta)$	$\Gamma = c(\Theta)$
$\text{Bin}(m, \theta)$	$(0, 1)$	$\log \frac{\theta}{1-\theta}$	$\mathbb{R}$
$\text{Poi}(\theta)$	$(0, \infty)$	$\log \theta$	$\mathbb{R}$
$\text{Exp}(\theta)$	$(0, \infty)$	$-1/\theta$	$(-\infty, 0)$
Power series	$(0, b)$	$\log \theta$	$(-\infty, \log b)$

In each case Γ contains an open interval, which is all the criterion asks.

3.1 Multiparameter version

$$f(x | \theta) = p(\theta)h(x) \exp \left\{ \sum_{j=1}^k c_j(\theta) T_j(x) \right\}, \quad \Gamma = \{(c_1(\theta), \dots, c_k(\theta)) : \theta \in \Theta\} \subseteq \mathbb{R}^k.$$

For n iid observations the sufficient statistic is $S = (\sum_i T_1(X_i), \dots, \sum_i T_k(X_i))$, and for completeness Γ must contain a **k -dimensional rectangle with non-empty interior**.

$N(\mu, \sigma^2)$:

$$f = \underbrace{\frac{e^{-\mu^2/2\sigma^2}}{\sqrt{2\pi\sigma^2}}}_{p(\theta)} \cdot \underbrace{1}_h \cdot \exp \left\{ \underbrace{\frac{\mu}{\sigma^2}}_{c_1} \underbrace{x}_{T_1} + \underbrace{\left(-\frac{1}{2\sigma^2}\right)}_{c_2} \underbrace{x^2}_{T_2} \right\}, \quad \Gamma = \mathbb{R} \times (-\infty, 0).$$

An open half-plane, so it contains a rectangle: $(\sum X_i, \sum X_i^2)$ is complete sufficient.

Gamma (α, β) (shape, scale, both unknown): $T = (\log x, x)$, $c = (\alpha - 1, -1/\beta)$, $\Gamma = (-1, \infty) \times (-\infty, 0)$. Complete.

Beta (θ, ϕ) : $T = (\log x, \log(1 - x))$, $c = (\theta - 1, \phi - 1)$, $\Gamma = (-1, \infty)^2$. Complete, with

$$S = \left(\sum_i \log X_i, \sum_i \log(1 - X_i) \right) \equiv \left(\prod_i X_i, \prod_i (1 - X_i) \right).$$

That is why the class notes record the Beta sufficient statistic as a pair of products: sums of logs are logs of products.

4 Curved families: where the criterion bites

Full rank versus curved

Write k for the number of (c_j, T_j) terms and d for the dimension of Θ .

- If $d = k$ and Γ has non-empty interior in $\mathbb{R}^k$, the family is **full rank** and the criterion applies: S is complete.
- If $d < k$, the natural parameters are constrained to a lower-dimensional surface. The family is **curved**, Γ contains no rectangle, and **the criterion says nothing**.

The standard example: $N(\theta, \theta^2)$, $\theta > 0$. Here $k = 2$ but $d = 1$:

$$(\eta_1, \eta_2) = \left(\frac{\theta}{\theta^2}, -\frac{1}{2\theta^2} \right) = \left(\frac{1}{\theta}, -\frac{1}{2\theta^2} \right), \quad \text{so} \quad \eta_2 = -\frac{1}{2}\eta_1^2:$$

a *parabola* in $\mathbb{R}^2$, which contains no rectangle. The criterion is unavailable — and in this case completeness genuinely fails.

Proof that $(\sum X_i, \sum X_i^2)$ is not complete for $N(\theta, \theta^2)$

Exhibit two *different* unbiased estimators of θ^2 , both functions of S . Using $\mathbb{E}X_i^2 = \text{Var} + \text{mean}^2 = \theta^2 + \theta^2 = 2\theta^2$ and $\mathbb{E}\bar{X}^2 = \frac{\theta^2}{n} + \theta^2 = \frac{(n+1)\theta^2}{n}$:

$$\mathbb{E}\left[\frac{1}{2n} \sum_i X_i^2\right] = \theta^2, \quad \mathbb{E}\left[\frac{n}{n+1} \bar{X}^2\right] = \theta^2.$$

So $g(S) = \frac{1}{2n} \sum_i X_i^2 - \frac{n}{n+1} \bar{X}^2$ satisfies $\mathbb{E}_\theta g(S) = 0$ for every θ , while $g \not\equiv 0$. That is a non-trivial zero function: **S is not complete**. (Simulation at $\theta = 2$, $n = 6$: $\mathbb{E}g = -0.001$ against a standard deviation of 1.61 — mean zero, but very far from the zero function.)

Consequence: Lehmann–Scheffé is unavailable here, and neither estimator above is a UMVUE.

5 The five consequences — why the course cares

This is the payoff. **One structure, five theorems.**

1. Sufficiency of fixed dimension

For $X_1, \dots, X_n$ iid,

$$f(\mathbf{x} \mid \theta) = p(\theta)^n \left(\prod_i h(x_i) \right) \exp \left\{ c(\theta) \sum_i T(x_i) \right\},$$

so by the factorisation theorem $S = \sum_i T(X_i)$ is sufficient — **one number, whatever n is**. Contrast the Cauchy, where you must keep all n order statistics. Data reduction of this kind is exactly what exponential families are for.

2. Completeness, hence Lehmann–Scheffé

If Γ contains an open interval (rectangle in the k -parameter case), S is **complete**. Combined with (1), Lehmann–Scheffé applies: *any* unbiased function of S is automatically *the* UMVUE. That is why the UMVUE recipe is so short for these families — find one unbiased $h(S)$ by series matching, and stop.

3. The Cramér–Rao bound is attained — for the right target

In natural form, $\log f = \log h(x) + \eta T(x) - A(\eta)$, so the score is

$$\frac{\partial}{\partial \eta} \log f = T(x) - A'(\eta) = T(x) - \mathbb{E}_\eta[T],$$

which is exactly the factorisation $k(\eta)[T - \psi(\eta)]$ required for equality in Cauchy–Schwarz. **So T attains the CRLB for $\psi(\eta) = \mathbb{E}_\eta[T]$** , and it is the UMVUE by (2).

The caveat that costs marks: this holds for $\mathbb{E}[T]$, **not for arbitrary ψ** . The UMVUE of $e^{-\lambda}$ in the Poisson (Mock Q3) is a function of S but is *not* S itself, and its variance sits strictly above the bound.

4. MLR, hence UMP tests

For $\theta_0 < \theta_1$,

$$\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = \frac{p(\theta_1)}{p(\theta_0)} \exp \{ [c(\theta_1) - c(\theta_0)] T(x) \},$$

increasing in $T(x)$ whenever c is non-decreasing. So the family has **MLR in T** , and Karlin–Rubin delivers a UMP test for every one-sided hypothesis. The Cauchy, not being an exponential family, has no MLR and no UMP test.

5. Finite-dimensional conjugate priors

$\prod_i f(x_i | \theta)$ depends on the data only through $(n, \sum_i T(x_i))$ — two numbers. So the family of priors

$$\pi(\theta) \propto p(\theta)^a \exp\{c(\theta)b\}, \quad (a, b) \text{ hyperparameters,}$$

is closed under sampling, with the update $(a, b) \mapsto (a + n, b + \sum_i T(x_i))$ — *pure arithmetic*. That is Beta–Binomial, Gamma–Poisson and Normal–Normal, all at once; look back at those update rules and you will see exactly this shape.

For the Cauchy no such compression exists, so the conjugate family must grow by one hyperparameter per observation — useless, which is the point of the Lecture 7 discussion.

	Exponential family	Cauchy / $U[0, \theta]$
Sufficient statistic	$\sum_i T(X_i)$, fixed dimension	order statistic (Cauchy); $X_{(n)}$ ($U[0, \theta]$)
Complete?	yes, if Γ has interior	no (Cauchy); yes but proved directly ($U[0, \theta]$)
CRLB	attained for $\mathbb{E}[T]$	regularity fails ($U[0, \theta]$)
MLR / UMP	yes, if c monotone	no (Cauchy); yes, directly ($U[0, \theta]$)
Conjugate prior	two hyperparameters	grows with n (Cauchy); Pareto ($U[0, \theta]$)

Common confusion: “So nothing works outside exponential families?”

Not so — and $U[0, \theta]$ is the proof. It is *not* an exponential family, yet $X_{(n)}$ is complete sufficient, the family has MLR, a UMP test exists (even a two-sided one), and there is a Pareto conjugate prior.

The correct statement is: being an exponential family is a **sufficient** condition for these conclusions, not a necessary one. When the hypothesis fails you simply lose the general theorem and must verify the conclusion *by hand* — which is precisely what the $U[0, \theta]$ questions in this course make you do.

6 The cumulant trick: moments for free

The identity

In natural form $f(x | \eta) = h(x)e^{\eta T(x) - A(\eta)}$,

$$\mathbb{E}_\eta[T] = A'(\eta), \quad \text{Var}_\eta(T) = A''(\eta), \quad I(\eta) = A''(\eta).$$

Proof. Integrating the density to 1 gives $\int h(x)e^{\eta T(x)} dx = e^{A(\eta)}$. Differentiate in η and divide by $e^{A(\eta)}$:

$$\int T(x)h(x)e^{\eta T(x) - A(\eta)} dx = A'(\eta) \implies \mathbb{E}[T] = A'(\eta).$$

Differentiate once more: $\mathbb{E}[T^2] = A'' + (A')^2$, so $\text{Var}(T) = A''(\eta)$. And since the score is $T - A'(\eta)$, the Fisher information is $\text{Var}(T) = A''(\eta)$. $\square$

Poisson, as a check. $\eta = \log \lambda$, $T = x$, $h = 1/x!$, and $A(\eta) = \lambda = e^\eta$. Then $\mathbb{E}[X] = A' = e^\eta = \lambda$ and $\text{Var}(X) = A'' = e^\eta = \lambda$ — both classical facts in one line, with no sums.

This subsumes the power-series identities of HW1 Q4

For a power-series family, $\eta = \log \theta$ (so $\frac{d\theta}{d\eta} = \theta$) and $A(\eta) = \log \tilde{A}(e^\eta)$ where $\tilde{A}(\theta) = \sum_x a_x \theta^x$. Then

$$\mathbb{E}[X] = \frac{dA}{d\eta} = \frac{\tilde{A}'(\theta)}{\tilde{A}(\theta)} \cdot \theta = \psi(\theta), \quad \text{Var}(X) = \frac{d^2 A}{d\eta^2} = \frac{d\psi}{d\eta} = \frac{d\psi}{d\theta} \cdot \frac{d\theta}{d\eta} = \theta \psi'(\theta).$$

Those are exactly the two boxed identities from Homework 1 Q4 — $\mathbb{E}X = \theta \tilde{A}'/\tilde{A}$ and $\text{Var} X = \theta \psi'(\theta)$ — obtained here with no computation at all. And the CRLB check falls out too: $\text{CRLB} = [\psi']^2/I = [\psi']^2\theta/\psi' = \theta\psi' = \text{Var}(X)$, attained. $\checkmark$

7 Master table

Family	$T(x)$	$c(\theta)$	Γ	Complete suff. stat.
Bern/Bin(m, θ)	x	$\log \frac{\theta}{1-\theta}$	$\mathbb{R}$	$\sum X_i$
Poi(θ)	x	$\log \theta$	$\mathbb{R}$	$\sum X_i$
Geom(θ)	x	$\log(1 - \theta)$	$(-\infty, 0)$	$\sum X_i$
NegBin(α, θ)	x	$\log(1 - \theta)$	$(-\infty, 0)$	$\sum X_i$
Power series	x	$\log \theta$	$(-\infty, \log b)$	$\sum X_i$
Exp(θ)	x	$-1/\theta$	$(-\infty, 0)$	$\sum X_i$
$N(\theta, 1)$	x	θ	$\mathbb{R}$	$\bar{X}$
$N(\mu, \sigma^2)$	(x, x^2)	$(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2})$	$\mathbb{R} \times (-\infty, 0)$	$(\sum X_i, \sum X_i^2)$
$N(0, \sigma^2)$	x^2	$-\frac{1}{2\sigma^2}$	$(-\infty, 0)$	$\sum X_i^2$
Gamma(α, β)	$(\log x, x)$	$(\alpha - 1, -\frac{1}{\beta})$	open rectangle	$(\sum \log X_i, \sum X_i)$
Beta(θ, ϕ)	$(\log x, \log(1-x))$	$(\theta - 1, \phi - 1)$	$(-1, \infty)^2$	$(\prod X_i, \prod(1 - X_i))$

Family	$T(x)$	$c(\theta)$	Γ	Complete suff. stat.
Multinomial	$(y_1, \dots, y_{k-1})$	$\log \frac{p_j}{1-p_1-\dots}$	$\mathbb{R}^{k-1}$	$(Y_1, \dots, Y_{k-1})$

Not exponential families

$N(\theta, \theta^2)$	curved: Γ is the parabola $\eta_2 = -\frac{1}{2}\eta_1^2$; sufficient but <i>not</i> complete
$U[0, \theta], U[\theta, \theta+1]$	support depends on θ ; handle by hand
Cauchy(θ)	mixed partial does not factor; sufficient statistic grows with n

8 Practice, with answers

- P1.** $X_1, \dots, X_n \stackrel{\text{iid}}{\sim}$ Geometric, $f(x | \theta) = \theta(1 - \theta)^x$, $x = 0, 1, 2, \dots$. Show this is an exponential family, identify Γ , and deduce that $\sum X_i$ is complete sufficient. Then find the UMVUE of θ for $n = 1$.
- P2.** Show that $N(\theta, \theta^2)$ is a curved exponential family, and exhibit a non-trivial zero function to show its sufficient statistic is not complete.
- P3.** $X_1, \dots, X_n \stackrel{\text{iid}}{\sim}$ Gamma(α, β) with both parameters unknown. Identify T , c , and Γ , and state the complete sufficient statistic.
- P4.** Use $\mathbb{E}[T] = A'(\eta)$, $\text{Var}(T) = A''(\eta)$ to obtain the mean and variance of the Binomial from its cumulant function.
- P5.** $U[0, \theta]$ is not an exponential family. Which conclusions of Section 5 nevertheless hold for it, and what does that tell you about the logical status of the theory?

Answers

P1. $f = \theta \exp\{x \log(1 - \theta)\}$, so $p = \theta$, $h = 1$, $T = x$, $c = \log(1 - \theta)$. Support $\{0, 1, 2, \dots\}$ is free of θ . ✓ As θ ranges over $(0, 1)$, $\Gamma = (-\infty, 0)$ — an open interval, so $\sum X_i$ is complete sufficient.

For $n = 1$, unbiasedness reads $\sum_x T(x)\theta(1 - \theta)^x = \theta$, i.e. with $p = 1 - \theta$, $\sum_x T(x)p^x = 1$ for all $p \in (0, 1)$. Matching coefficients: $T(0) = 1$ and $T(x) = 0$ for $x \geq 1$. So the UMVUE is $\mathbb{I}\{X = 0\}$, with variance $\theta(1 - \theta)$. (Sensible: you see one observation, and the only unbiased signal about $\theta = \mathbb{P}(X = 0)$ is whether X was 0.)

P2. With $\mu = \theta$, $\sigma^2 = \theta^2$, $(\eta_1, \eta_2) = (1/\theta, -1/(2\theta^2))$, so $\eta_2 = -\frac{1}{2}\eta_1^2$: the natural parameters lie on a parabola, a one-dimensional curve in $\mathbb{R}^2$. It contains no rectangle, so the completeness criterion does not apply — the family is *curved*.

Completeness genuinely fails. Since $\mathbb{E}X_i^2 = 2\theta^2$ and $\mathbb{E}\bar{X}^2 = \frac{(n+1)}{n}\theta^2$, both $\frac{1}{2n}\sum_i X_i^2$ and $\frac{n}{n+1}\bar{X}^2$ are unbiased for θ^2 , so

$$g(S) = \frac{1}{2n} \sum_i X_i^2 - \frac{n}{n+1} \bar{X}^2$$

has $\mathbb{E}_\theta g = 0$ for all θ with $g \not\equiv 0$. Hence S is not complete.

P3. $f = \frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha} = \frac{1}{\Gamma(\alpha)\beta^\alpha} \exp\{(\alpha-1)\log x - \frac{x}{\beta}\}$, so $T = (\log x, x)$ and $c = (\alpha-1, -1/\beta)$, giving $\Gamma = (-1, \infty) \times (-\infty, 0)$ — an open rectangle. Therefore $S = (\sum_i \log X_i, \sum_i X_i)$ is complete sufficient.

P4. $X \sim \text{Bin}(m, \theta)$ has $\eta = \log \frac{\theta}{1-\theta}$, so $\theta = \frac{e^\eta}{1+e^\eta}$ and, from $p(\theta) = (1 - \theta)^m$,

$$A(\eta) = -\log p = -m \log(1 - \theta) = m \log(1 + e^\eta).$$

Then

$$\mathbb{E}[X] = A'(\eta) = \frac{me^\eta}{1 + e^\eta} = m\theta, \quad \text{Var}(X) = A''(\eta) = \frac{me^\eta}{(1 + e^\eta)^2} = m\theta(1 - \theta). \quad \checkmark$$

P5. It fails condition (i) — the support $[0, \theta]$ moves. Yet: $X_{(n)}$ is sufficient (factorisation) *and* complete (proved directly by differentiating $\int_0^\theta g(u)nu^{n-1}du = 0$ in θ); the family has MLR in $X_{(n)}$ (the ratio is constant then jumps to $+\infty$); a UMP test exists, and unusually a two-sided one as well; and the Pareto family is conjugate. What genuinely breaks is Cramér–Rao, which needs the regularity that moving support destroys.

Moral: the exponential-family hypothesis is *sufficient* for the good behaviour, not *necessary*. Losing it means losing the general theorem, so you must verify the conclusion by hand — which is exactly why $U[0, \theta]$ appears so often in this course.